ANN ASSIGNMENT

Digit Recognition with LeNet-5 on MNIST

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GitHub repository: ann-assignment

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1 Dataset

The dataset used in this project is MNIST, a standard benchmark for handwritten digit recognition. The program loads it through `torchvision.datasets.MNIST` with separate training and test splits. In the code, the dataset contains 70,000 images in total:

- 60,000 training images
- 10,000 test images

Each image belongs to one of 10 classes, representing digits 0 through 9. Every sample is a 28×28 grayscale image, which the program converts into a 1×28×28 tensor for the CNN. If flattened, each image contains 784 pixel values.

Table 1: MNIST dataset summary

Property	Value
Total images	70,000
Training images	60,000
Test images	10,000
Number of classes	10
Image size	28×28 grayscale
Program input tensor	1×28×28
Flattened pixel values	784

2 LeNet-5 Model

LeNet-5 is one of the earliest and most influential convolutional neural network architectures for digit classification. In this project, it is adapted for MNIST by using one grayscale input channel and a padded first convolution so the spatial size remains 28×28 before pooling.

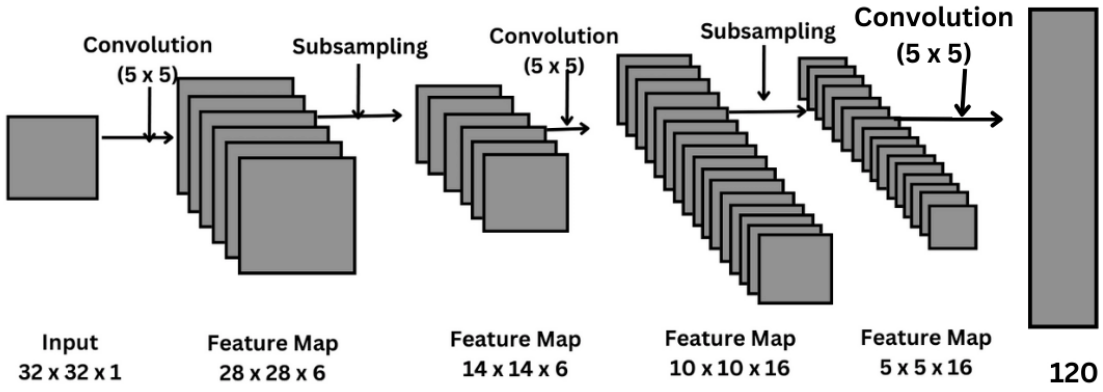

Figure 1: LeNet-5 CNN architecture reference used for the project report.

Table 2: Model architecture used in the project

Layer	Configuration / Output	Parameters
Input	1×28×28 grayscale image tensor	0
Conv1	6 filters, 5×5 kernel, padding 2, Tanh; output 6×28×28	156
Pool1	Average pooling, 2×2, stride 2; output 6×14×14	0
Conv2	16 filters, 5×5 kernel, Tanh; output 16×10×10	2,416
Pool2	Average pooling, 2×2, stride 2; output 16×5×5	0
Conv3	120 filters, 5×5 kernel, Tanh; output 120×1×1	48,120
Flatten	Reshape to a 120-dimensional vector	0
FC1	Linear 120→84, Tanh; output 84	10,164
Output	Linear 84→10 logits, one per digit class	850

The network has 61,706 trainable parameters. It is trained with Adam, batch size 64, learning rate 0.001, weight decay 1e-4, and a step learning-rate scheduler.

3 Observations

1. The model classified the clearer digits very confidently, with several predictions close to 100% confidence.
2. Digits 4 and 6 were more difficult because their handwriting styles were less distinct than the simpler digits.
3. The model confused 4 with 8 and 9 with 7, which shows that similar stroke patterns can lead to misclassification.
4. The preprocessing pipeline is important: grayscale conversion, inversion, cropping, padding, resizing, and normalization make custom images more consistent with MNIST.
5. Even with a compact size of 61,706 parameters, LeNet-5 achieved strong performance on handwritten digit recognition, which makes it a good lightweight CNN baseline.

The custom image files are named by their true labels, so 0.png corresponds to class 0, 1.png to class 1, and so on.

Table 3: Inference results on custom images

Image	Actual class	Predicted class	Confidence
0.png	0	0	99.99%
1.png	1	1	99.99%
2.png	2	2	99.89%
3.png	3	3	100.00%
4.png	4	8	62.25%
5.png	5	5	99.98%
6.png	6	6	73.87%
7.png	7	7	98.86%
8.png	8	8	100.00%
9.png	9	7	99.95%

4 Conclusion

This project demonstrates that a LeNet-5 style convolutional neural network is effective for MNIST digit recognition. The dataset is simple enough for a compact model, yet it still reveals where ambiguity in handwriting can challenge the classifier. The results show that the model is highly reliable on most digits, while a few visually similar samples remain difficult cases.